Hyunchang Kweon

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Education

Seoul National University Expected Feb 2027
B.S. in Aerospace Engineering Seoul, Korea
GPA: 3.59 / 4.3

Purdue University Spring 2026
Exchange Student – Aeronautical and Astronautical Engineering West Lafayette, IN, USA
Semester GPA: 3.93 / 4.00 (Dean's List & Semester Honors)

Experience

SNU Navigation and Electronic Systems Lab (NESL) Sep 2025 – Present
Undergraduate Mentee – Advisor: Prof. Chan Gook Park Seoul, Korea

- Investigating semantic segmentation and its application to UAV control as part of a bachelor's thesis.

SNU Learning and Decision Systems Lab (LDS) Jun 2025 – Aug 2025
Undergraduate Intern – Advisor: Prof. Insoon Yang Seoul, Korea

- Independently studied physics-informed neural networks and reinforcement learning during a summer research internship.

Republic of Korea Navy, Naval Academy Jan 2023 – Sep 2024
Military Translator / Interpreter Jinhae, Korea

- Assisted Naval Academy midshipmen with military English courses and supported wargame simulations.
- Provided translation and interpreting services for military officers at international events, including the Freedom Shield Joint Military Exercise, ROKNA Joint Cruise Training, Combined Military Exercise Talisman Sabre, and the LIMA Malaysia Maritime Defense Exhibition.

Selected Projects

Occlusion-Aware UAV Tracking Apr 2026
Course Project, Purdue University West Lafayette, IN, USA

- Built a Pygame simulation for aerial tracking of moving targets around obstacles.
- Developed a model predictive controller using sequential least squares programming (SLSQP), analytical gradients, and collision-avoidance constraints to maintain target distance and line of sight.
- Implemented evaluation scripts and compared three target-trajectory predictors using visibility, tracking error, control effort, and computation time as evaluation metrics.

Aerial 3D Reconstruction and Semantic Understanding Apr 2026
Course Project, Purdue University West Lafayette, IN, USA

- Collaborated on evaluating 3D Gaussian Splatting, CLIP, and SAM 2 on UAVScenes to assess trade-offs between reconstruction quality and semantic accuracy across flight altitudes.
- Evaluated FFT-based motion-blur filtering on one UAVScenes sequence, improving held-out reconstruction PSNR by 1.1 dB over the unfiltered baseline at the best tested threshold.

Selected Projects (continued)

23rd Korea Robotic Aircraft Competition Apr 2025 – Sep 2025
Control Software Engineering Team Seoul, Korea

- Developed a model aircraft with vertical takeoff and landing (VTOL) capabilities and visual target recognition for autonomous rescue.
- Developed perception and control software using YOLO and ROS 2 for target recognition and alignment control.
- Team placed among the top 7 of 37 teams.

2025 AIAA Design Build Fly Competition Dec 2024 – Apr 2025
CFD Team Leader Seoul, Korea & Tucson, USA

- Led the CFD team, responsible for airfoil selection and analysis of flight characteristics using automated Python scripts built with the Ansys Fluid API.
- Conducted fluid–structure interaction analysis with Ansys Mechanical.

Aerospace Capstone Design, J.Marple Aug 2024 – Nov 2024
Team Leader Seoul, Korea

- Developed a MATLAB/Simulink model of the Crazyflie PID controller firmware as part of an industry–academia collaboration with J.Marple.
- Performed software-in-the-loop simulations (SILS) to verify the recreated model.

Leadership / Extracurricular Activities

Bulnabi – Drone Club at SNU Aug 2024 – Present
President (Jan 2025 – Dec 2025) Seoul, Korea
Executive Team Member (Aug 2024 – Dec 2024)

- Organized seminars on 3D printing, soldering, drone assembly, and programming, led by experienced members, to help newcomers contribute to club projects.
- Helped organize the club's participation in the annual aerospace exhibition hosted by SNU's Department of Aerospace Engineering.

CARDIO Aerospace Drone Winter Camp Dec 2024 – Jan 2025
Team Member Seoul, Korea

- Completed a winter training program provided by the Korea Aerospace Research Institute and Korea Aeromodels Association on UAV and radio-controlled aircraft development.

SounDreaM, Songwriting Club at SNU Jun 2022 – Present
Treasurer / Drummer Seoul, Korea

Technical Skills and Languages

Programming	Python, C, PyTorch, R
Robotics & Simulation	ROS 2, Gazebo, PX4 Autopilot, MATLAB/Simulink
CAD & Engineering Analysis	SolidWorks, Fusion 360, Ansys, OpenVSP
Other Tools	Linux, LaTeX
Languages	Korean (native); English (fluent, TOEFL 119/120); Chinese (advanced, HSK Level 6)